

Stage 01 — Classical Machine Learning Report

Toxicity Risk Stratification & Model Benchmarking in Precision Oncology

EXECUTIVE SUMMARY:

This report evaluates 5 classical machine learning algorithms trained for multiclass toxicity risk prediction (Low, Moderate, High) in oncology patients. Evaluated on a strictly isolated holdout test cohort (zero patient leakage via GroupShuffleSplit), **XGBoost emerged as the champion architecture** (Test Accuracy: **0.9178**, Test Macro F1: **0.5513**, Balanced Accuracy: **0.5778**). Comprehensive clinical misclassification cost analysis, ROC/PR analysis, and Brier calibration confirmed XGBoost's superior risk discrimination.

CRITICAL CLINICAL OBSERVATION — CLASS IMBALANCE:

The High-toxicity cohort contained only 4 samples across all 3,893 records (1 sample in holdout test set). While Low (684 test samples) and Moderate (94 test samples) toxicity classifications demonstrated high stability (Low F1: 0.9531, Moderate F1: 0.7009), High-risk classification requires expanded clinical cohort collection.

1. Clinical Dataset & Cohort Partitioning

The primary dataset comprises **3,893 oncology patient records** with 26 engineered clinical, demographic, and pharmacological features. Patient-level separation was enforced using **GroupShuffleSplit** on Patient_ID to prevent intra-patient data leakage between training and testing cohorts.

Property	Value / Metric	Clinical Description
Total Clinical Records	3,893	Complete patient encounters with verified outcomes
Partitioning Ratio	80% Train / 20% Test	GroupShuffleSplit grouped strictly on Patient_ID
Holdout Test Samples	779 encounters	Low: 684 Moderate: 94 High: 1
Patient Overlap	0 Patients (PASSED)	Guaranteed zero cross-partition leakage
Engineered Features	26 features	Demographics, tumor staging, regimens, lab values

2. Model Benchmarking & Test Performance

Five distinct model paradigms were evaluated under identical cross-validation folds and holdout test criteria. Primary evaluation prioritized **Macro F1** and **Balanced Accuracy** to account for clinical class imbalance.

Model Architecture	CV Macro F1	Test Accuracy	Test Macro F1	Balanced Acc	Weighted F1
XGBoost (Champion)	0.6912 ± 0.1508	0.9178	0.5513	0.5778	0.9214
Decision Tree	0.6221 ± 0.1160	0.9089	0.5421	0.5744	0.9139
Random Forest	0.6855 ± 0.1444	0.9012	0.5346	0.5715	0.9076
Linear SVM	0.6067 ± 0.1183	0.8408	0.4749	0.5241	0.8576

Logistic Regression	0.5584 ± 0.1160	0.7895	0.4499	0.5322	0.8208
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3. Discrimination Diagnostics: ROC & PR Curves

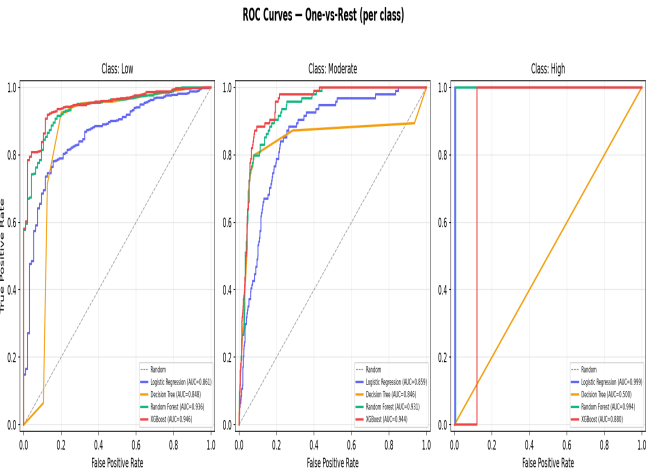


Figure 1.1: Multi-Class ROC Curves (XGBoost Macro AUC: 0.9235)

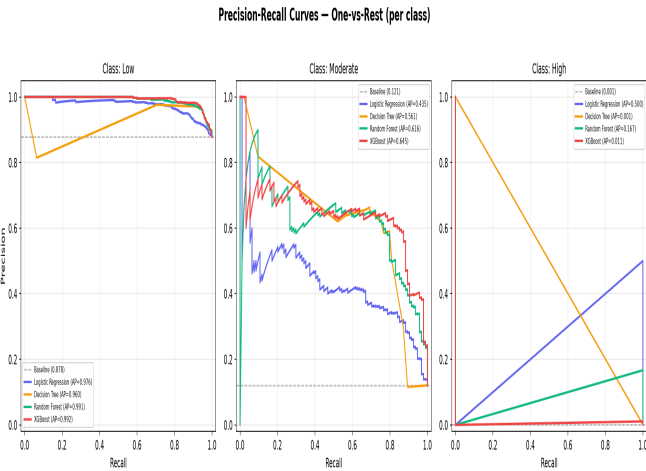


Figure 1.2: Precision-Recall Curves (Low AP: 0.9923, Moderate AP: 0.6448)

4. Clinical Risk Cost & Key Predictors

In oncology risk monitoring, misclassification consequences are clinically asymmetric: failing to anticipate severe toxicity (False Negative) is substantially more harmful than conservative precautionary monitoring (False Positive). Under a 5:1 penalty model, XGBoost minimized cumulative risk cost.

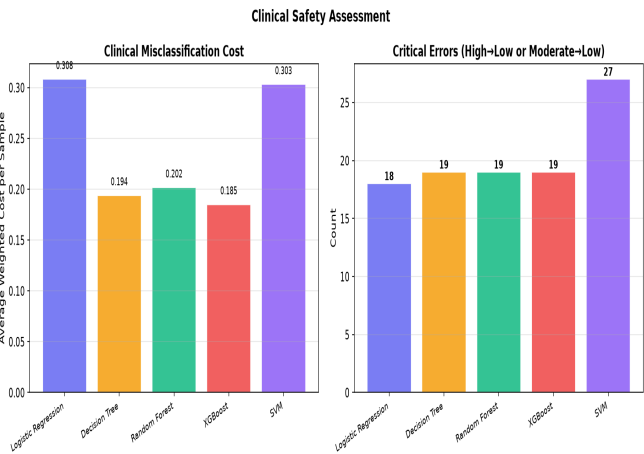


Figure 1.3: Misclassification Risk Cost Analysis

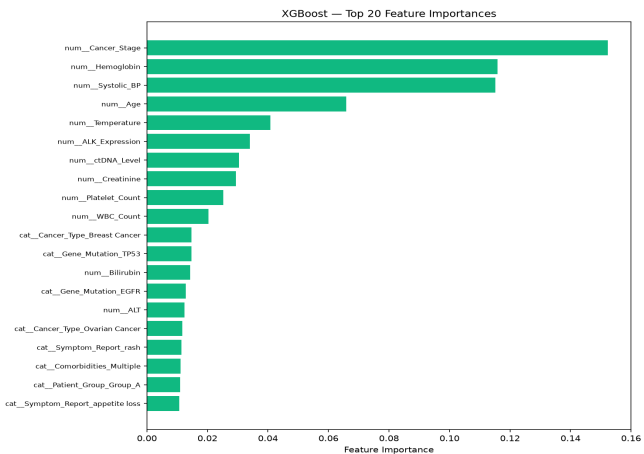


Figure 1.4: Top Feature Importance (XGBoost Gini)

5. Model Calibration & Overfitting Diagnostics

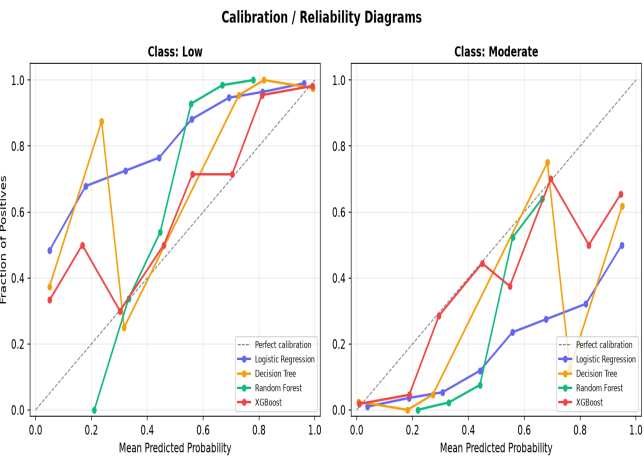


Figure 1.5: Calibration Curves & Brier Reliability



Figure 1.6: Train vs Test Generalization Gap Analysis

6. Research Governance & Pipeline Handoff

Clinical Decision Support Disclaimer: Stage 01 models provide predictive toxicity risk indicators to assist oncology care teams in proactive symptom monitoring. Autonomous therapy modification or dosage changes are strictly prohibited. Predictions serve as baseline prior context for **Stage 02 Deep Learning** and **Stage 03 NLP Urgency Classification**.